CONCEPT DOCUMENT & TECHNICAL WHITE PAPER | STAGE: FUNCTIONAL MVP | PRIMARY JURISDICTION: HYDERABAD (GHMC / HMDA)

SITEGEN

Parametric Architectural Feasibility & Land Acquisition Intelligence Engine

Transforming weeks of fragmented pre-acquisition due diligence into an instant, multi-dimensional parametric simulation answering: "Should I buy this plot of land?"

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1. Executive Summary

The strategic necessity for instant land acquisition feasibility intelligence

In real estate development and land acquisition, billions of dollars are committed annually under severe informational asymmetry. Determining whether a specific land parcel is viable currently requires manual intervention across siloed domains: interpreting complex municipal zoning byelaws, calculating regulatory setback buffers, hand-drafting CAD massing models, and constructing speculative financial spreadsheets. This traditional feasibility cycle takes 2 to 4 weeks and costs substantial consulting capital per plot, creating severe friction in fast-moving land markets.

SiteGen eliminates this friction through a unified parametric computation engine. By synthesizing GIS parcel geometries, dynamic municipal regulatory frameworks (GHMC/HMDA), automated 3D buildable envelope calculations, and real-time development financial models, SiteGen compresses a multi-week due diligence study into a sub-second, interactive browser session.

The result is complete acquisition intelligence: developers, architects, and institutional funds can evaluate dozens of spatial scenarios, verify zoning adherence, optimize Floor Area Ratio (FAR), stress-test development margins, and generate executive investment memos in minutes before committing capital.

2. Core Value Pillars

Three fundamental pillars powering SiteGen

SPEED & AGILITY

Instant Parametric Math

Replaces manual CAD modeling with mathematical setback offsets, automatic FAR utilization calculations, and live geometric massing updates at 60 FPS.

REGULATORY RIGOR

Coded Municipal Logic

Directly encodes road-width stepped tables, setback formulas, height caps, and parking ratio requirements specific to Hyderabad (GHMC & HMDA).

FINANCIAL CERTAINTY

Unified Underwriting

Directly bridges physical massing to Indian real-estate financial KPIs: Total Development Cost, Land Cost Ratios, Revenue, Net Profit, and ROI Margins.

THE SITEGEN WORKFLOW PARADIGM:

Land Polygon !' Zoning Byelaws !' Buildable Envelope !' 3D Massing !' F

One continuous, reactive feedback loop recalculating in under 10 milliseconds across every user parameter change.

3. The Core Problem Statement

Why traditional land feasibility analysis is dangerously inefficient

The Core Question: "Should I buy this plot of land?"

Every land transaction represents a high-stakes, irreversible capital expenditure. Yet buyers must evaluate plot potential under substantial informational friction, outdated regulatory rulebooks, and speculative volumetric estimates. The absence of an integrated computational engine forces developers into costly guesswork.

4. Key Failure Points in the Current Industry Workflow

The four critical friction barriers holding back rapid land acquisition

FRICTION POINT 01

Regulatory Fragmentation & Opacity

Municipal zoning codes (e.g. GHMC/HMDA) link setback requirements and maximum FAR to road width tiers and boundary dimensions. Misjudging road width thresholds by just a few feet can restrict allowable height and reduce buildable saleable area by up to 35%.

FRICTION POINT 02

Disconnected Toolchains & Sunk Time

Evaluating a single plot requires GIS technicians, architects running CAD/Rhino, and financial analysts building Excel models. This disjointed process takes 10 to 20 days. In competitive land tenders, this latency results in lost deals or rushed, unvetted bids.

FRICTION POINT 03

Disconnected Financial Modeling

Spatial density and financial yield are fundamentally interlinked, yet traditionally analyzed in isolation. Pro-forma models often assume theoretical FAR numbers that the physical plot geometry and setback rules cannot geometrically accommodate.

FRICTION POINT 04

Lack of Multi-Scenario Exploration

Because architectural massing is labor-intensive, teams typically test only 1 or 2 building configurations. Developers miss optimal trade-offs between floor count, parking basement excavation depth, floor heights, and overall profit margins.

5. Market Context: Why Hyderabad First?

The strategic geographic beachhead for SiteGen

Hyderabad has emerged as one of India's fastest-growing real estate markets, characterized by rapid commercial IT corridor expansions (Gachibowli, HITEC City, Financial District, Kokapet, Neopolis) and expansive residential growth along the Outer Ring Road (ORR) and Regional Ring Road (RRR).

Key characteristics making Hyderabad the ideal validation testbed for SiteGen:

- Progressive High-Rise Policy: Telangana features flexible FAR frameworks coupled with stringent road-width and fire-fighting setback corridors, making parametric setback math critical.
- Extreme Land Price Escalation: High land acquisition costs demand strict margin optimization; even a 2% miscalculation translates into crores of rupees in capital variance.
- High Velocity of Transactions: High transaction volumes by institutional funds and tier-1 builders demand rapid pre-screening tools.

6. Product Architecture & Technical Implementation

A modern, modular reactive architecture engineered for 60 FPS performance

SYSTEM TOPOLOGY: CLIENT-SIDE REACTIVE COMPUTATION PIPELINE

STEP 1
GIS Map Engine
Leaflet / OSM
Geo-Coords

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STEP 2
Zoning Engine
GHMC Tables
Setback Offsets

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STEP 3
Geometric Core
Polygon Math
Buildable Footprint

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STEP 4
3D Massing
R3F / Three.js
SunCalc Shadows

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STEP 5
Underwriting
Capex / Opex
Executive PDF

Centralized state synchronization managed by Zustand with immutable parameter dispatching and zero-lag DOM recalculation.

7. Deep Dive: Key Functional Modules Built

Overview of the operational components delivered in the MVP

MODULE 1: GIS & SPATIAL INGESTION

Cartesian & Geodetic Boundary Engine

Integrates Leaflet and OpenStreetMap with a custom fallback layer. Translates latitude/longitude coordinates into local Cartesian coordinates (feet) around the centroid, computes polygon area, perimeter, and aspect ratio, and enables interactive site boundary adjustments.

MODULE 2: REGULATORY ENGINE

Hyderabad GHMC/HMDA Rules Translator

Encodes regulatory tables matching road widths (e.g., <20ft, 20-30ft, 30-40ft, 40-60ft, 60-80ft, >80ft). Dynamically computes front, rear, and side setback buffers, allowable FAR (up to 4.0), and maximum building height envelopes, while allowing expert manual overrides.

MODULE 3: 3D PARAMETRIC MASSING

WebGL Extrusions & SunCalc Solar Simulation

Renders dynamic 3D building masses via React Three Fiber. Generates floor slab demarcations, setbacks, and road baselines. Integrates SunCalc to accurately position astronomical directional sunlight and cast real-time shadows for Morning, Noon, and Evening solar analysis.

MODULE 4: FINANCIAL & CAPACITY ANALYTICS

Real Estate Pro-Forma & Sensitivity Modeler

Calculates Gross Floor Area (GFA), net saleable area (based on efficiency factor), and mandatory parking bays. Instantly projects Land Acquisition Cost, Construction Cost, Additional Approvals/Contingency Capex, Projected Gross Revenue, Net Profit, and ROI Margin.

MODULE 5: EXECUTIVE REPORTING

Vectorized Investment Feasibility PDF Generation

Exports complete, branded investment memos via jsPDF containing all dimensional, regulatory, massing, and financial metrics.

8. Target Stakeholders & Transformative Impact

Whose lives change and how SiteGen fundamentally reshapes real estate acquisition

PRIMARY USER GROUP

Real Estate Developers, Land Syndicates & Builders

- Impact: Cuts due diligence time from 20 days to 2 minutes during competitive land bidding.
- Value Created: Eliminates multi-crore capital misallocations on unbuildable plots; gives rigorous quantitative leverage during landowner price negotiations and

DESIGN PROFESSIONALS

Architectural Firms, Master Planners & Urban Designers

- Impact: Automates mundane setback offset calculations, height checks, and initial building massing.
- Value Created: Frees design talent from tedious pre-feasibility math; allows studios to deliver rapid, client-facing volumetric options and competitive bids with r

CAPITAL ALLOCATORS

Private Equity Real Estate Funds, NBFCs & Sovereign Wealth Funds

- Impact: Standardizes investment committee underwriting with automated, defensible feasibility reports.
- Value Created: Reduces risk by verifying that projected cashflows are rooted in true spatial reality rather than speculative spreadsheet formulas; accelerates d

CIVIC & URBAN ECOSYSTEM

Municipal Planning Authorities (GHMC / HMDA) & Urban Corridors

- Impact: Submissions pre-aligned with statutory road-width and setback codes reduce permit rejection rates.
- Value Created: Promotes healthier urban morphology with compliant setbacks, adequate fire corridors, realistic parking provisions, and solar exposure across

9. Commercial Model & Monetization Strategy

Scalable B2B SaaS business model targeting high-value enterprise accounts

PRO DEVELOPER

₹15,000 / mo

- Unlimited plot feasibility runs
- 3D massing & solar shadows
- Up to 20 exportable PDF reports
- Hyderabad jurisdiction access

ENTERPRISE FUND

₹75,000 / mo

- Multi-seat team workspace
- Unlimited PDF acquisition memos
- Custom financial pro-forma rules
- Direct PostGIS API access

INSTITUTIONAL API

Custom Pricing

- Headless programmatic API
- Batch cadastral plot analysis
- Bank collateral valuation engine
- SLA dedicated support

10. Current Stage of Development

Detailed breakdown of verified prototype capabilities vs planned enhancements

CURRENT OPERATIONAL STATUS

Stage: Functional Working Prototype (MVP)

SiteGen has successfully transitioned beyond the conceptual and wireframe phase into an interactive, browser-based working prototype. The core client-side mathematical engines (geometry, setbacks, building height, FAR constraints, solar positioning, and financial calculations) are fully implemented and running synchronously at zero latency.

11. Technical Implementation Status

Current capabilities versus planned pilot architecture

COMPONENT / CAPABILITY	TECHNICAL STACK	STATUS
Interactive Leaflet GIS Mapping	React Leaflet / OpenStreetMap	Completed & Live
Geometric Setback Offset Engine	Custom Polygon Offset Math (TS)	Completed & Live
Parametric 3D WebGL Massing	React Three Fiber / Three.js	Completed & Live
Astronomical Sun & Shadow Path	SunCalc / Real-time Dir Light	Completed & Live
Real-time Financial Pro-Forma	Zustand / Recharts / TS Engine	Completed & Live
Automated Feasibility PDF Report	jsPDF Client-side Generator	Completed & Live
PostGIS Spatial Registry Backend	Django REST / PostgreSQL PostGIS	Under Development
Multi-City Zoning Rule Expansion	Bengaluru (BDA) / Mumbai (BMC)	Roadmap (Q1 2027)

12. Strategic Product Roadmap

Phased plan from current MVP to institutional pilot and pan-India expansion

PHASE 1 (COMPLETED)

Core MVP Validation

- Parametric calculations
- Real-time 3D extrusion
- Hyderabad GHMC ruleset
- Integrated financial dashboard
- PDF memo export

PHASE 2 (MONTHS 1-4)

Cloud Backend & Pilot

- Django + PostGIS deployment
- Persistent user project accounts
- Multi-scenario comparison
- Pilot testing with 5 Hyderabad developers and architecture firms

PHASE 3 (MONTHS 5-12)

Scale & Cadastral Ingestion

- Pan-India regulatory engines (Bengaluru BDA, Mumbai BMC)
- Automated government land parcel registry overlay
- Institutional fund underwriting portal

PROJECT CONCLUSION & SUMMARY:

SiteGen bridges computational geometry, municipal law, and investment finance to eliminate guesswork from land acquisition. Currently operating as a high-fidelity prototype